

EG1311: Design & Make

Program Flow

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Program Flow

- Machine code (assembly) instruction set
 - Data Transfer (read and write memory)
 - Data Manipulation (change memory)
 - **Program Flow (change next execution)**
 - Branching (conditionals)
 - Repetition (loops)
 - Abstraction (functions)

Programming: Conditionals

- Change program flow based on Boolean

```
if (a < b) {  
    // do something  
} else {  
    // do something else  
}
```

Indentation

Programming: Loops

- Repeat lines

```
while (a < b) {  
    // do something  
    // repeatedly  
}
```

Programming: Loops

- Repeat lines

```
int i = 0;
while (i < 50) {
    // do something once
    i = i + 1;
}
```

Programming: Loops

- Repeat lines

```
int i = 0;
while (i < 50) {
    // do something once
    i++;
}
```

Programming: Loops

- Repeat lines

```
for (int i = 0; i < 50; i++) {  
    // do something once  
}
```

Practice

```
int a = 100;  
while ((a >= 0) || (a % 2)) {  
    a--;  
}  
int b = a;
```

What number
is stored in
variable b?

A

B

C

D

A) 1

B) 0

C) -1

D) -2

Programming: Functions

- Parametrically reusable code
- Inputs $\rightarrow$ one output

```
int add(int x, int y) {  
    return x + y;  
}  
  
int c = add(a, b);
```

Programming: Functions

- Parametrically reusable code
- Inputs $\rightarrow$ one output

```
void wait() {  
    // code to delay execution  
}
```

Programming: Best Practices

- Write detailed comments
- Be clear and simple (don't be cryptic)
- Use descriptive variable/function names
- Indent code properly, align curly braces
- Use variables instead of hard-coded values

Practice

```
char a = 0;  
for (char i = 120; i > 0; i++) {  
    a++;  
}  
char b = a;
```

What number
is stored in
variable b?

A

B

C

D

A) 7

B) 8

C) 9

D) infinite loop

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